



CHAPTER 3

PERIODIC

TABLE



3.1

Classification Of Elements

PERIODIC TABLE

- Metals (main-group)
- Metals (transition)
- Metals (inner transition)
- Metalloids
- Nonmetals

INNER TRANSITION ELEMENTS

6	Lanthanides	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
7	Actinides	90 Th 232.0	91 Pa (231)	92 U 238.0	93 Np (237)	94 Pu (242)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (260)

INTRODUCTION

The Periodic Table is a tabular display of the chemical elements, which are arranged by increasing proton number (also known as atomic number), electronic configuration, and recurring chemical properties. The structure of the table shows periodic trends. The **seven rows** of the table, **called periods**, generally have metals on the left and nonmetals on the right. The **columns, called groups**, contain elements with **similar chemical properties**.

Period

- ✓ **Seven horizontal** rows of elements, which are numbered from **1 to 7**.
- ✓ The elements are arranged in order of the **increasing proton number from left to right** of Periodic Table.
- ✓ The valence shells of elements in the **same period** have the **same principal quantum number, n** .

Group

- ✓ **Vertical** columns of elements in the Periodic Table.
- ✓ The groups in the Periodic Table are numbered from **1 to 18**.
- ✓ Elements placed in the same group, have
 - i. **similar chemical properties**
 - ii. **same number of valence electrons**

Main Groups in the Periodic Table

Group 1	Alkali metals (except H)
Group 2	Alkaline earth metals
Group 3 - Group 12	Transition metals
Group 16	Chalcogens
Group 17	Halogens
Group 18	Inert or noble gases

Blocks

Elements in the periodic table are classified into *s*, *p*, *d* and *f*-blocks according to the type of subshell being filled by the electron of highest energy of the elements.

1																	18	
1	1s	2															1s	
2	Li	Be											B	C	N	O	F	Ne
3	Na	Mg	3	4	5	6	7	8	9	10	11	12	Al	Si	P	S	Cl	Ar
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
6	Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
7	Fr	Ra		Rf	Db	Sg	Bh	Hs	Mt	Uun	Uuu	Uub		Uuq	Uuh			Uuo

La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

s block	Group 1 & 2 Valence electronic configuration: ns^1 to ns^2 Eg. $_{12}\text{Mg} = 1s^2 2s^2 2p^6 3s^2$ consists of alkaline metals and alkaline earth metal.
p block	Elements of Group 13 – 18 Valence electronic configuration: $ns^2 np^1$ to $ns^2 np^6$ Eg. $_{15}\text{P}: 1s^2 2s^2 2p^6 3s^2 3p^3$ consists of metals, metalloids & non-metals elements.
d block	Elements in Group 3 – 12 Consist of the transition elements . Valence electronic configuration: $ns^2 (n-1)d^1$ to $ns^2 (n-1)d^{10}$ Eg. $_{23}\text{V}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$
f block	Elements in Series of lanthanides (4f) and actinides (5f). Final electron filled into a <i>f</i> subshell. <i>f</i> -block elements are mostly radioactive.

Metals

✓ Properties: Dense, malleable, ductile, shinny, good conductor of heat and electricity.

metals

1																	18	
1	H																	He
2	Li	Be											B	C	N	O	F	Ne
3	Na	Mg	3	4	5	6	7	8	9	10	11	12	Al	Si	P	S	Cl	Ar
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
6	Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
7	Fr	Ra		Rf	Db	Sg	Bh	Hs	Mt	Uun	Uuu	Uub		Uuq		Uuh		Uuo
			La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	
			Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	

Metalloids

✓ Known as semi-metals and have properties between those of metals and nonmetals tend to be semiconductors and widely used in manufacturing sector.

metalloids																				
1	2													13	14	15	16	17	18	
1	H													B	C	N	O	F	He	
2	Li	Be													Al	Si	P	S	Cl	Ne
3	Na	Mg	3	4	5	6	7	8	9	10	11	12	Al	Si	P	S	Cl	Ar		
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr		
5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe		
6	Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn		
7	Fr	Ra		Rf	Db	Sg	Bh	Hs	Mt	Uun	Uuu	Uub		Uuq		Uuh		Uuo		

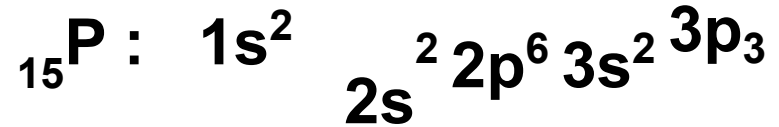
Nonmetals

- ✓ Further across the period towards the right, elements gradually lose their metallic character and gained nonmetallic features. Generally, gases which are poor conductor.

nonmetals																				
1	2													13	14	15	16	17	18	
1	H													B	C	N	O	F	He	
2	Li	Be													5	6	7	8	9	10
3	Na	Mg	3	4	5	6	7	8	9	10	11	12	Al	Si	P	S	Cl	Ar		
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr		
5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe		
6	Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn		
7	Fr	Ra		Rf	Db	Sg	Bh	Hs	Mt	Uun	Uuu	Uub		Uuq		Uuh		Uuo		

Determining Position of Elements Based On Electronic Configuration

Example: Phosphorus

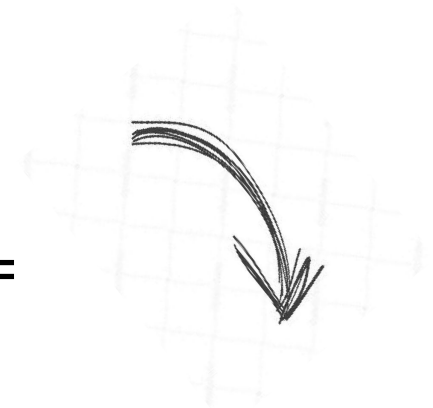


Period: Highest principal quantum number : **3**

Block: Highest occupied subshell : **p**

Number of valence electrons : $2 + 3 = \mathbf{5}$

Group: *p* block (10 + number of valence electrons) : $10 + 5 = \mathbf{15}$

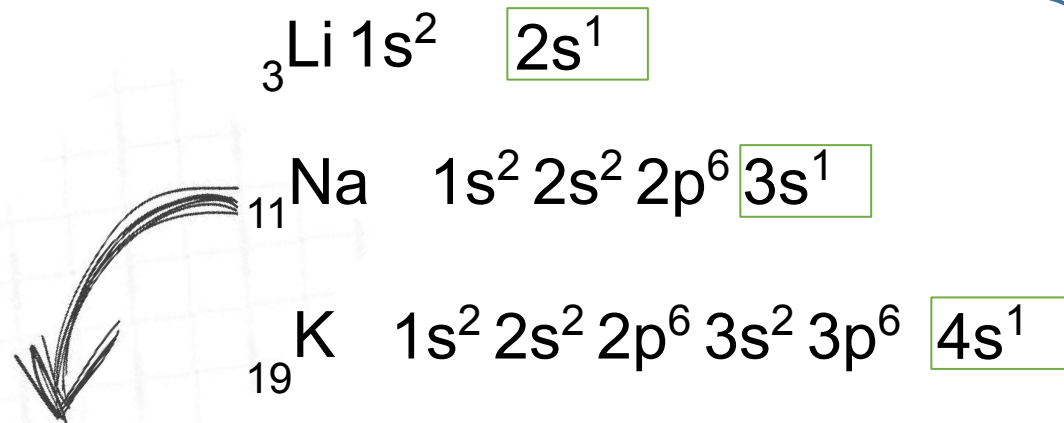


Period 3,
p block
Group 15

For main group elements (s block and p block):

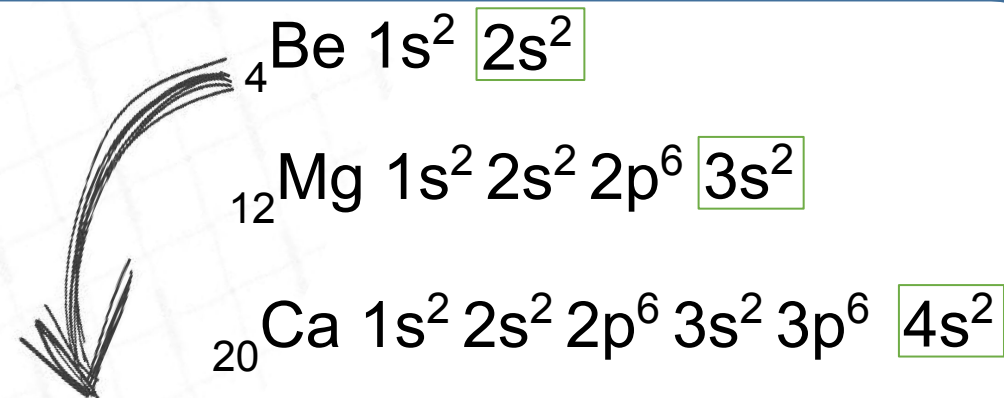
- ✓ valence electrons are those in the **outermost shell** or electron shell with the **highest principal quantum number, n**.
- ✓ the **outermost shell** is also known as **valence shell**.

Example 1:



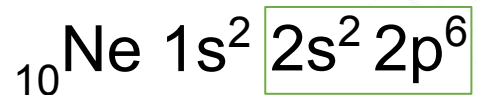
- ✓ Highest principal quantum number : 3 (**Period 3**)
- ✓ Occupied by **only 1 $\bar{e}$** .
- ✓ Hence, number of valence $\bar{e}$ = 1 (**Group 1**).
- ✓ **Valence electronic configuration: $3s^1$**

Example 2:



- ✓ Highest principal quantum number : 2 (**Period 2**).
- ✓ Occupied by **2 $\bar{e}$** .
- ✓ Hence, number of valence $\bar{e}$ = 2 (**Group 2**).
- ✓ **Valence electronic configuration: $2s^2$**

Example 3:

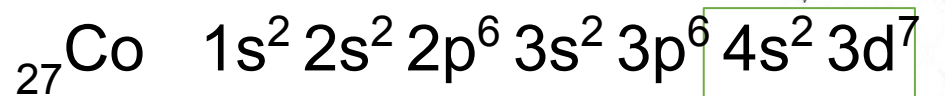


- The highest principal quantum number : 2 (**Period 2**).
- **Electron of highest energy is filled into p subshell (p block).**
- Number of valence $\bar{e} = 8$ (**Group 18**).
- **Valence electronic configuration: $2s^2 2p^6$.**

For **d-block elements**,

- ✓ valence electrons are **not only the electrons in the outermost shell (n), but also include those in the $(n-1)d$ subshell.**

Example 4:



- The highest principal quantum number : 4 (**Period 4**)
- **Electron of highest energy is filled into 3d subshell (d block).**
- Number of valence $\bar{e} = 9$ (**Group 9**).
- **Valence electronic configuration: $4s^2 3d^7$**

PAST YEAR ASSIGNMENT QUESTION

QUESTION 1

TABLE 1 shows the proton number of five elements

TABLE 1

Elements	A	B	C	D	E
Proton number	17	16	11	8	18

Based on **TABLE 1**,

(i) Which element is located in Group 1 and Period 3?

Element C

Number of valence
electrons = 1

Highest principal
quantum number = 3

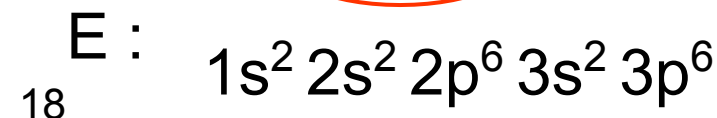
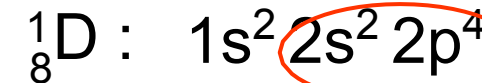
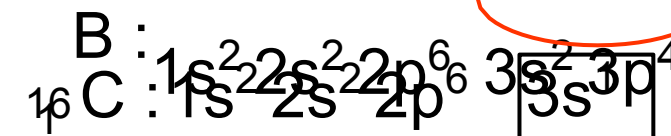
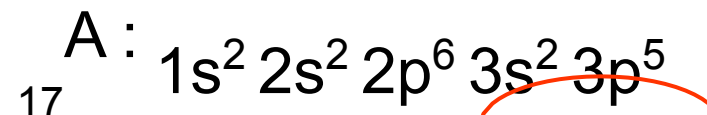
(ii) Which elements belong to the same group?

Element B and D

same number of valence electrons

Note :

Electronic configuration





3.2

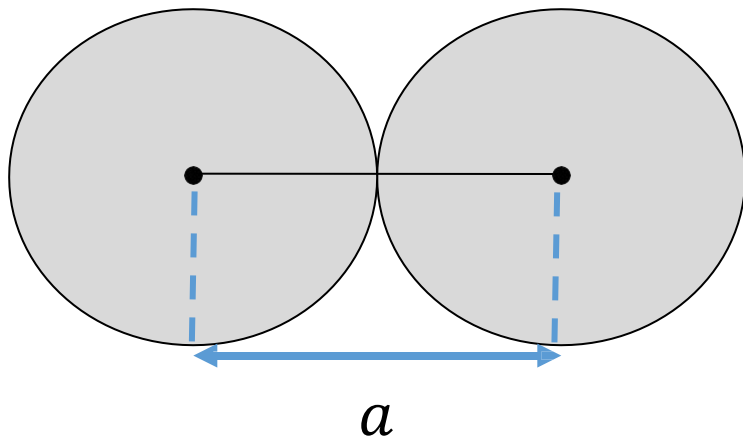
Periodicity

PERIODICITY

- ✓ Periodicity is the periodic trend in properties of elements.

Periodic Trends In The Size Of Atom (Atomic Radius)

- ✓ The size @ radius of atom is difficult to be defined exactly because the electron cloud has no clear boundary.
- ✓ To solve this, distance between the 2 nuclei in a molecule is determined.
- ✓ Radius, r = half of the distance between the nuclei of two adjacent identical atoms.



$$\text{Radius, } r = \frac{a}{2}$$

Periodic Trends In The Size Of Atom (Atomic Radius)

Factors that determine atomic radii are:

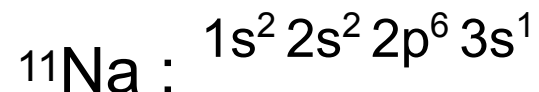
1. Effective nuclear charge (Z_{eff})

The net positive charge experienced by an electron in an atom. It is not the full nuclear charge because there is some shielding of the nucleus by the other electrons in the atom.

2. The Shielding @ Screening Effect

is the result of the mutual repulsion between the electrons in the inner shell with those in the outermost shell.

Example



10 inner electrons ($1s^2 2s^2 2p^6$) shield the nucleus from the valence electrons (-10)

$$Z_{\text{eff}} = Z - S$$

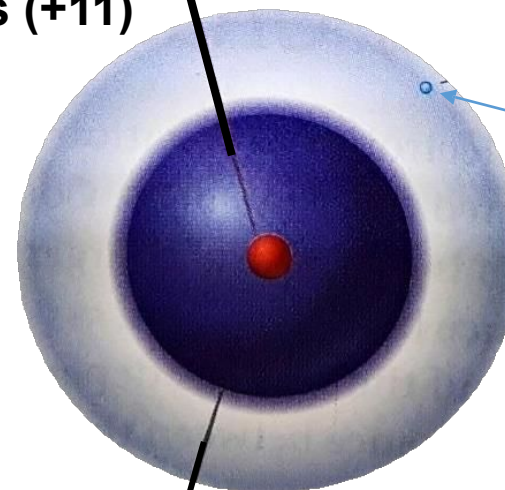
Z = proton number @ nuclear charge

S = shielding factor

(= number of inner electrons)

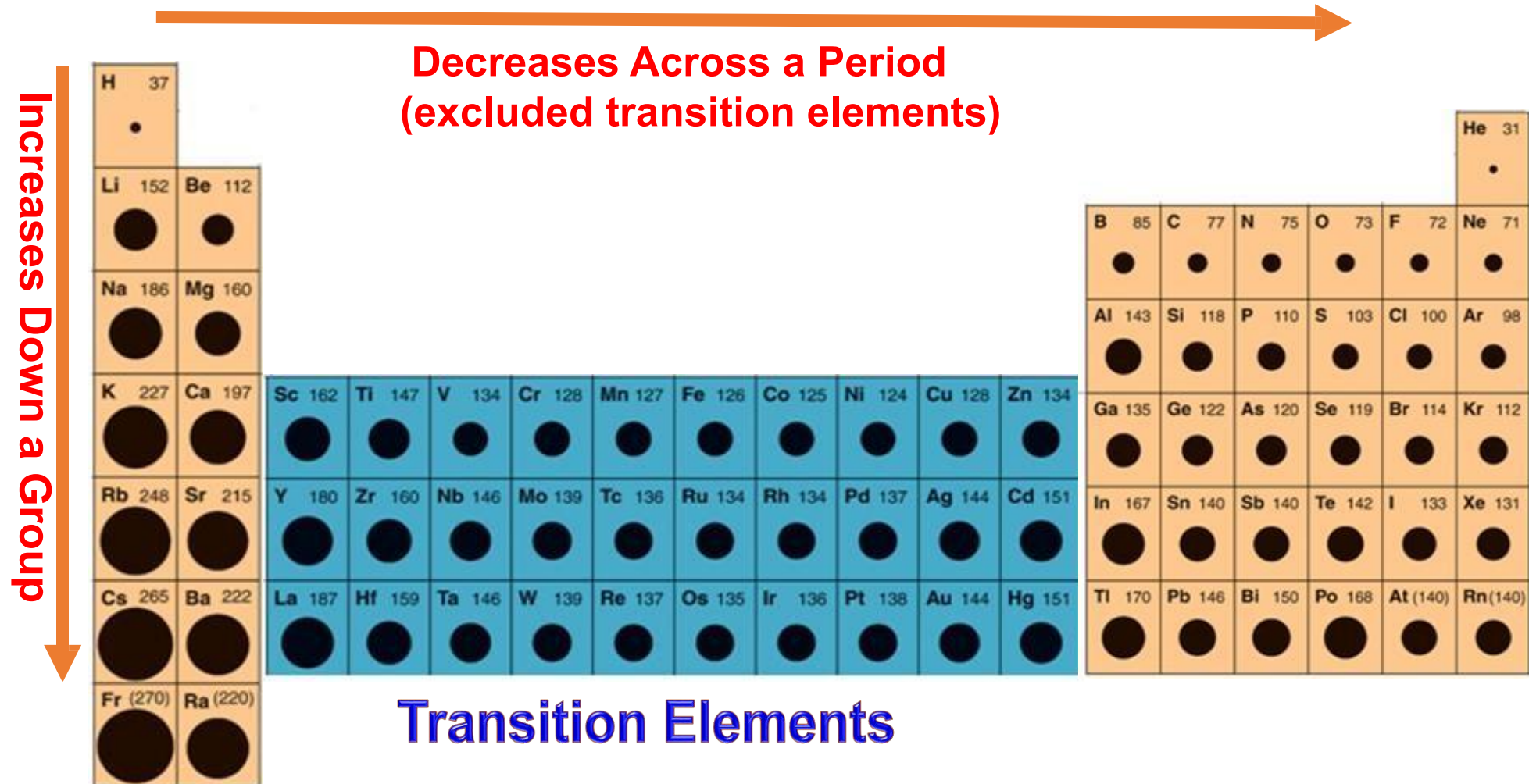
Sodium nucleus
contains 11
protons (+11)

$$\begin{aligned} Z_{\text{eff}} &= Z - S \\ &= 11 - 10 \\ &= +1 \end{aligned}$$



Overall Periodic Trends in Atomic Radius

- 1. Decreases across a period** from left to right.
- 2. Increases going down a group.**
- 3. Relatively constant across the first series of transition metals.**



1. Across Period 3

- ✓ As the **proton number increases**, **effective nuclear charge, Z_{eff}** increases.
- ✓ The **attraction between the nucleus and the valence e^-** becomes stronger.
- ✓ As a result, **valence electrons are pulled closer to the nucleus**, the **atomic radius @ size decreases**

ELEMENT	Z	Electronic Configuration	S	Z_{eff}	Atomic Radius (pm)
Na	11	$1s^2 2s^2 2p^6 3s^1$	10	+1	186
Mg	12	$1s^2 2s^2 2p^6 3s^2$	10	+2	160
Al	13	$1s^2 2s^2 2p^6 3s^2 3p^1$	10	+3	143
Si	14	$1s^2 2s^2 2p^6 3s^2 3p^2$	10	+4	117
P	15	$1s^2 2s^2 2p^6 3s^2 3p^3$	10	+5	110
S	16	$1s^2 2s^2 2p^6 3s^2 3p^4$	10	+6	104
Cl	17	$1s^2 2s^2 2p^6 3s^2 3p^5$	10	+7	99
Ar	18	$1s^2 2s^2 2p^6 3s^2 3p^6$	10	+8	94

Left to right



PAST YEAR ASSIGNMENT QUESTION

QUESTION 2

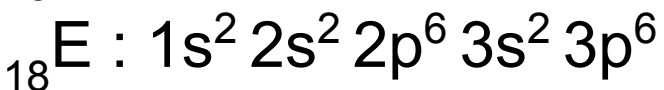
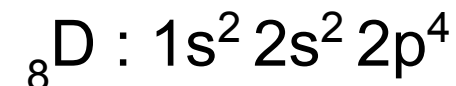
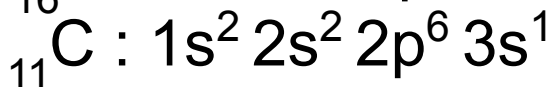
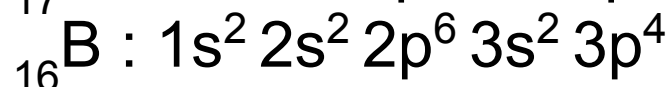
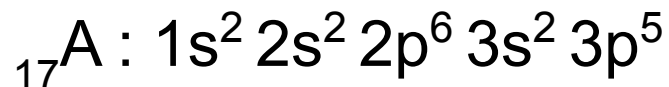
TABLE 1 shows the proton number of five elements

TABLE 1

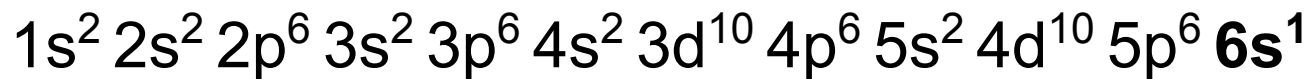
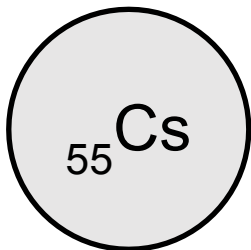
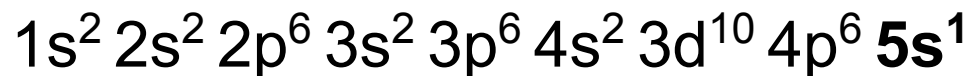
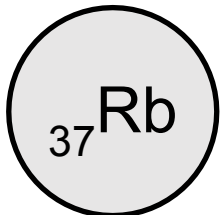
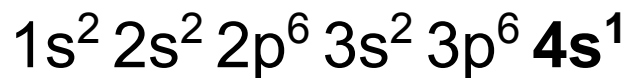
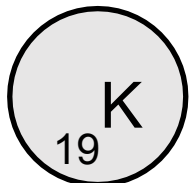
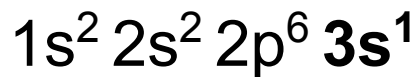
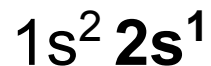
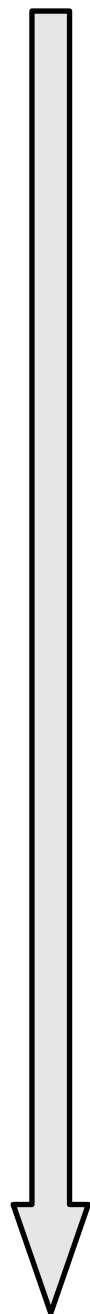
Elements	A	B	C	D	E
Proton number	17	16	11	8	18

(a) Arrange these elements in order of decreasing atomic radius. Briefly explain your answer

ANSWER



- **C ,B , A and E are elements of the same period.**
- **From C, B, A to E, the proton number increases.**
- **The effective nuclear charge increases.**
- **Thus, the attraction between nucleus and electrons become valence stronger.**
- **Therefore, the atomic radius decreases from C, B, A to E.**
- **D is the smallest due to D has lower principal quantum @ less number of electron shells than C,B,A and E.**
- **D experiences lower shielding effect which results in stronger attraction between nucleus and valence electrons.**



2. Down Group 1

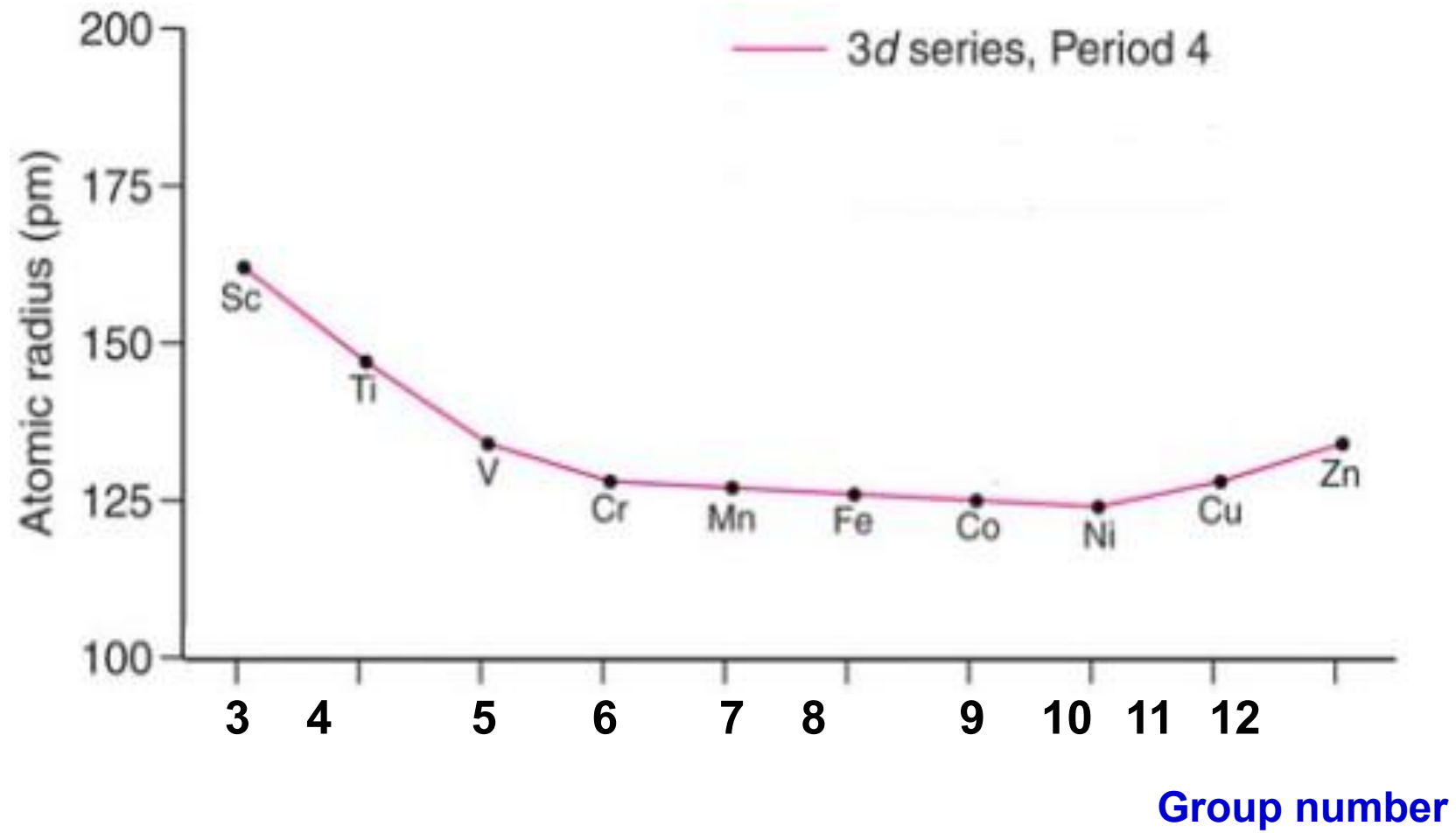
- ✓ As the number of electron shells increases, shielding effect increases.
- ✓ The attraction between the nucleus and the valence electrons become weaker.
- ✓ As a result, atomic radius @ size increases from Li to Fr.

3. Across First Row Of Transition Elements

- ✓ The transition metals are the elements which have incompletely filled d subshells.
- ✓ Across the period of the first row of transition metal, the atomic radii do not change significantly.
- ✓ As proton number increases, the electron will be added to the inner 3d subshell, but the number of electrons in outermost 4s subshell remains the same.
- ✓ **The increase in nuclear charge is cancelled by the increase in shielding effect of the inner electrons.**
- ✓ The **attraction** between nucleus and outermost electrons is **almost the same** throughout the series.
- ✓ Thus, there is **no significant change in atomic radius** of the transition metals.

Element	Atomic radius (pm)	Z	Electronic configuration	$Z_{\text{eff}} = Z - S$
Mn	127	25	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$	$25 - 23 = +2$
Fe	126	26	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$	$26 - 24 = +2$
Co	125	27	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$	$27 - 25 = +2$
Ni	124	28	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$	$28 - 26 = +2$

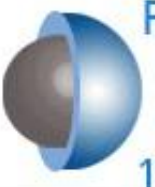
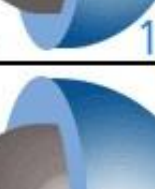
- ✓ The table above illustrates how the increase in nuclear charge is offset by the increase in shielding effect. Atomic radius @ size of the first row of transition metals are relatively constant.



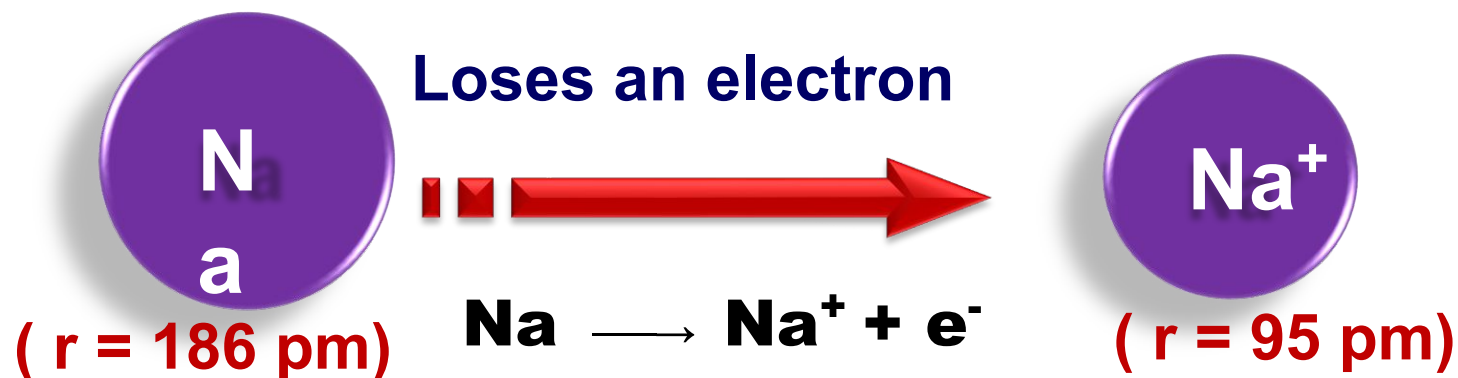
Size Of Atoms And Their Corresponding Ions

In general,

- ✓ a **cation** is **smaller** than its **atom**.
- ✓ while an **anion** is **larger** than its **atom**.

Group 1	Group 2	Group 13	Group 16	Group 17
Li^+  Li 90 134	Be^{2+}  Be 59 90	B^{3+}  B 41 82	O  O^{2-} 73 126	F  F^- 71 119
Na^+  Na 116 154	Mg^{2+}  Mg 86 130	Al^{3+}  Al 68 118	S  S^{2-} 102 170	Cl  Cl^- 99 167
K^+  K 152 196	Ca^{2+}  Ca 114 174	Ga^{3+}  Ga 76 126	Se  Se^{2-} 116 184	Br  Br^- 114 182
Rb^+  Rb 166 211	Sr^{2+}  Sr 132 192	In^{3+}  In 94 144	Te  Te^{2-} 135 207	I  I^- 133 206
atomic radius (pm)				

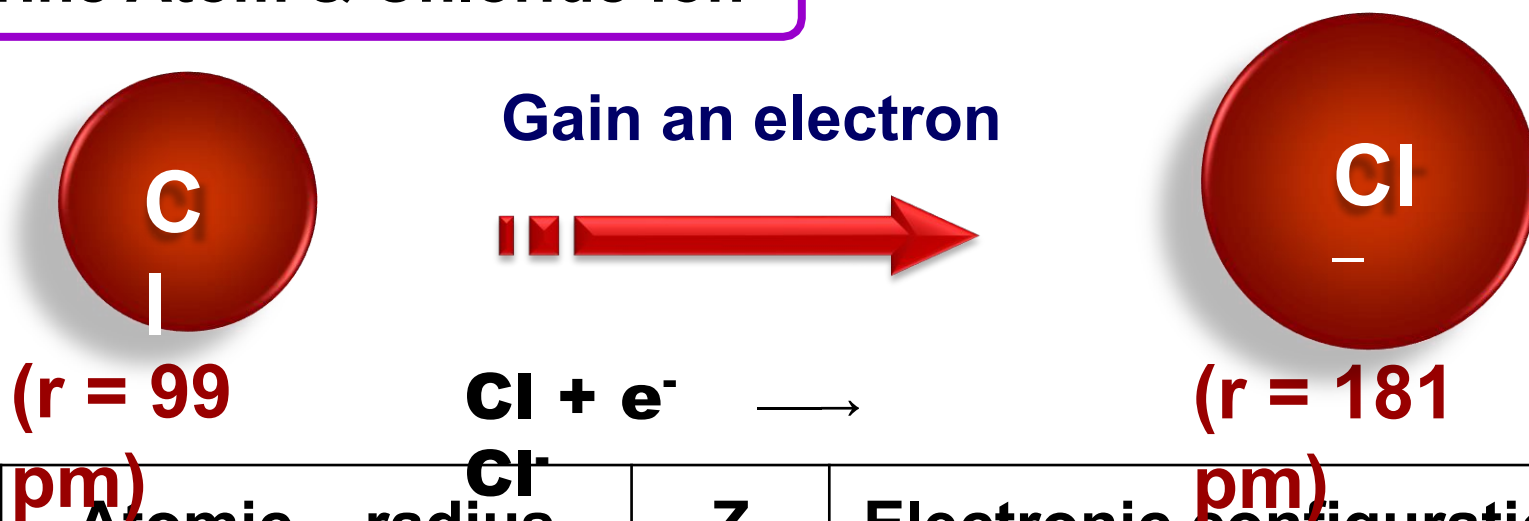
Size of Sodium Atom & Sodium ion



Element	Atomic radius (pm)	Z	Electronic configuration
Na	186	11	$1s^2 2s^2 2p^6 3s^1$
Na ⁺	95	11	$1s^2 2s^2 2p^6$

- ✓ Na atom **loses @ donates electron** to form Na⁺ ion.
- ✓ As the electrons is removed from Na atom, the **number of electron shells decreases** hence the **shielding effect decreases**.
- ✓ Since Na⁺ ion has fewer electrons than Na atom, the **electron-electron repulsion decreases**.
- ✓ The **attraction of the nucleus towards the remaining electron becomes stronger**.
- ✓ Hence **Na⁺ ion is smaller** than Na atom.

Size of Chlorine Atom & Chloride ion



Element	Atomic radius (pm)	Z	Electronic configuration
Cl	99	17	$1s^2 2s^2 2p^6 3s^2 3p^5$
Cl^-	181	17	$1s^2 2s^2 2p^6 3s^2 3p^6$

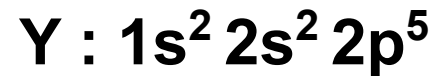
- ✓ Cl atom gains @ accepts electron to form Cl^- ion.
- ✓ Since Cl^- ion has more electrons than Cl atom, the electron-electron repulsion increases.
- ✓ The attraction of the nucleus towards the valence electrons becomes weaker
- ✓ Hence Cl^- ion is **bigger** than Cl atom.

PAST YEAR ASSIGNMENT QUESTION

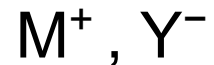
QUESTION 3

- (a) Compound ***MY*** is formed when ***M*** transfers an electron to ***Y***. ***M*** and ***Y*** are the elements with proton number of 3 and 9.
- i) State the possible ions formed from atoms ***M*** and ***Y***.
Write the electronic configuration of ions in part (i).

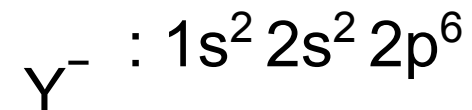
ANSWER



Possible ions formed :



Electronic configuration:

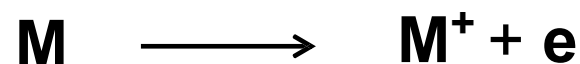


PAST YEAR ASSIGNMENT QUESTION

QUESTION 3

- (a) Compound ***MY*** is formed when ***M*** transfers an electron to ***Y***. ***M*** and ***Y*** are the elements with proton number of 3 and 9.
- ii) Explain the differences in size between the atom and its corresponding ion for ***M*** and ***Y***.

ANSWER



- **M loses 1 electron** to form **M⁺ ion**.
- As the electrons is removed from M atom, the **number of electron shells decreases** hence the **shielding effect decreases**.
- Since M⁺ ion has fewer electrons than M atom, the **electron-electron repulsion decreases**.
- The **attraction of the nucleus towards the remaining electron becomes stronger**.
- Hence **M⁺ ion is smaller** than M atom.

PAST YEAR ASSIGNMENT QUESTION

QUESTION 3

- (a) Compound ***MY*** is formed when ***M*** transfers an electron to ***Y***. ***M*** and ***Y*** are the elements with proton number of 3 and 9.
- ii) Explain the differences in size between the atom and its corresponding ion for ***M*** and ***Y***.

ANSWER



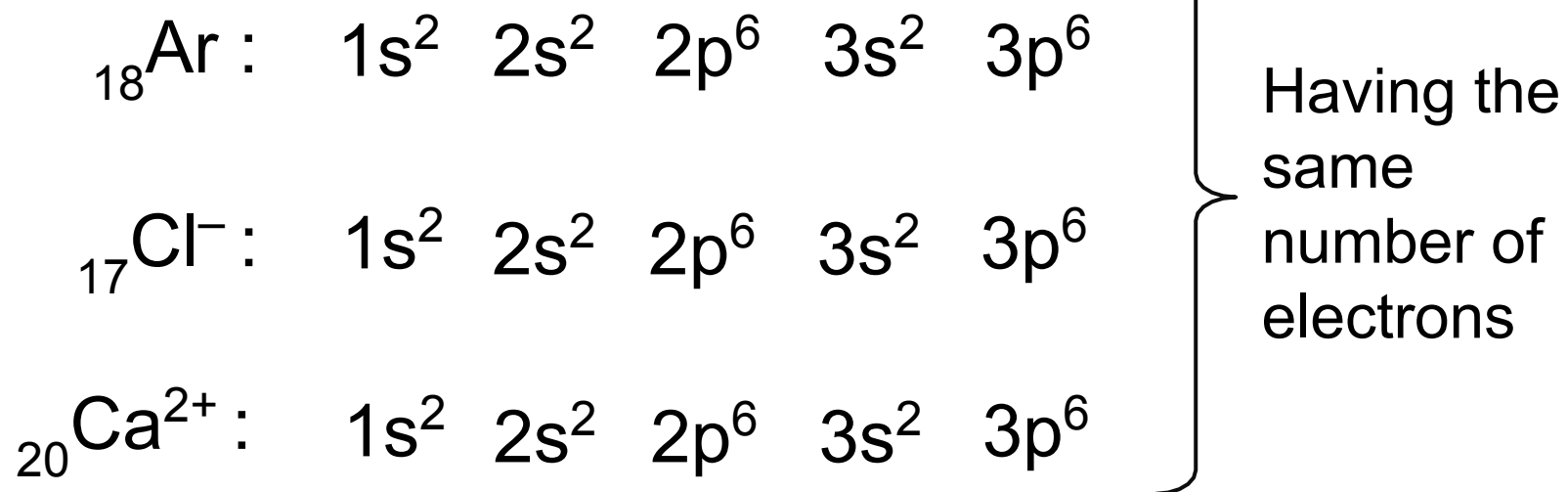
- **Y gains 1 electron to form Y^{-} .**
- **Since Y^{-} ion has more electrons than Y, the electron–electron repulsion increases.**
- **The attraction of the nucleus towards the valence electrons becomes weaker**
- **Hence Y^{-} ion is bigger than Y atom.**

Isoelectronic

Isoelectronic species

⇒ species that have the same number of electrons or the same electronic configuration.

Example:

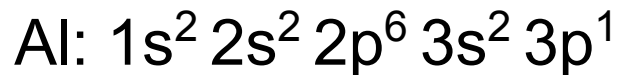
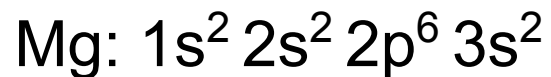


Compare The Radius Of Isoelectronic Species

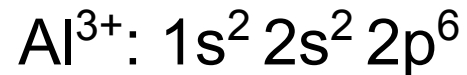
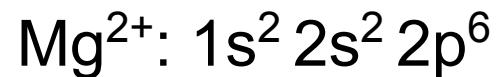
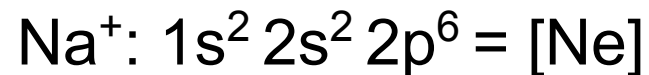
- ✓ Na, Mg and Al are elements located in **Period 3**.
- ✓ The formation of cations from neutral atom of main group elements occur when one or more electrons are removed from the outermost shell to achieve **stable electronic configuration**.

Example:

Neutral atoms



Cations

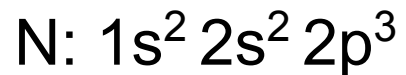
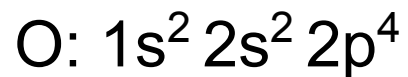
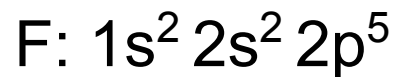


Compare The Radius Of Isoelectronic Species

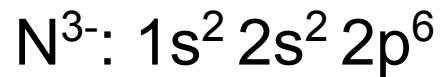
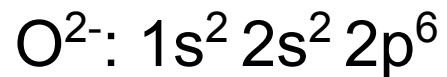
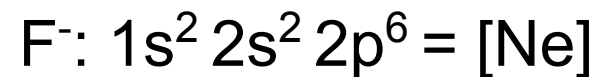
- ✓ F, O and N are period 2 elements .
- ✓ The formation of anions from neutral atom of main group elements occur when one or more electrons are added to the outermost shell to achieve stable electronic configuration.

Example:

Neutral atoms



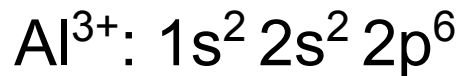
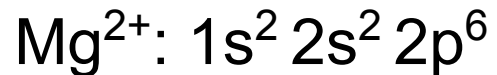
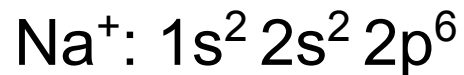
Anions



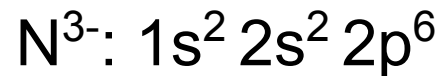
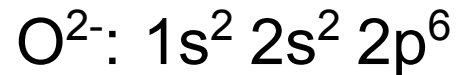
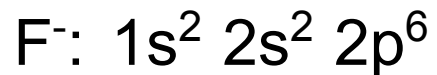
Compare The Radius Of Isoelectronic Species

Example:

Cations



Anions



- ✓ All the cations and anions above are known as **isoelectronic ions** because they have **same electronic configuration**. Figure 3.2.1 shows some of the series of isoelectronic ions.

GROUP

1

2

13

15

16

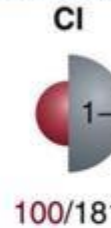
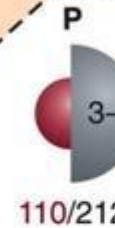
17

2



All ions have 10 electrons

3



4

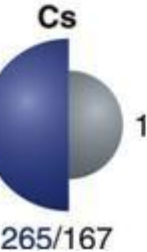


5



A series of isoelectronic ions

6

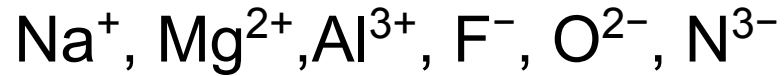


Series of isoelectronic ions

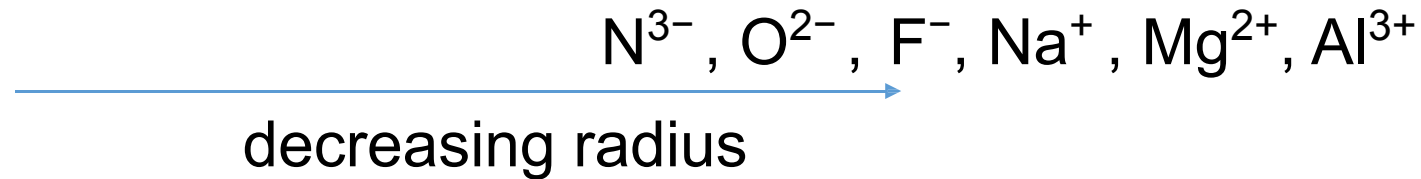
Example:**Variation in Radius of Isoelectronic Species**

Ion	N^{3-}	O^{2-}	F^{-}	Na^{+}	Mg^{2+}	Al^{3+}
No. of electrons	10	10	10	10	10	10
Same no. of electrons →						
Z @ Nuclear charge	+7	+8	+9	+11	+12	+13
Increasing nuclear charge →						
S @ number of inner e ⁻	2	2	2	2	2	2
Z _{eff}	+5	+6	+7	+9	+10	+11
Increasing Z _{eff} →						
Radius(Å)	1.71	1.40	1.33	0.97	0.66	0.51
Radius DECREASES						

Arrange the following ions in order of decreasing radius. Explain.



ANSWER



- ✓ N^{3-} , O^{2-} , F^- , Na^+ , Mg^{2+} and Al^{3+} are isoelectronic species with electronic configuration $1s^2 2s^2 2p^6$.
- ✓ As proton number increases, **effective nuclear charge increase** from N^{3-} to Al^{3+} .
- ✓ The **attraction of nucleus towards valence electrons** becomes **stronger**.
- ✓ Therefore, **ionic radius decreases** from N^{3-} to Al^{3+} .

Example:

	${}^{15}\text{P}^{3-}$	${}^{16}\text{S}^{2-}$	${}^{17}\text{Cl}^{-}$
No. of proton	15	16	17
No. of electron	18	18	18

$1s^2 2s^2 2p^6 3s^2 3p^6$

Predict the size of the isoelectronic species. Explain.

✓ From P^{3-} to Cl^{-} , ionic size decreases.

Explanation:

✓ As proton number increases from P^{3-} to Cl^{-} , effective nuclear charge, Z_{eff} increases which lead to a **greater attraction between the nucleus and the valence electrons**, hence **reduce the size of ion**.

Variation In The Ionic Radii Across Period 2 And Period 3

Variation Of Ionic Radii Across Period 3

Proton Number

Ion

Electronic

configuration

11

Na⁺

1s² 2s² 2p⁶

12

Mg²⁺

1s² 2s² 2p⁶

13

Al³⁺

1s² 2s² 2p⁶

14

Si⁴⁺

1s² 2s² 2p⁶

15

P³⁻

1s² 2s² 2p⁶ 3s² 3p⁶

16

S²⁻

1s² 2s² 2p⁶ 3s² 3p⁶

17

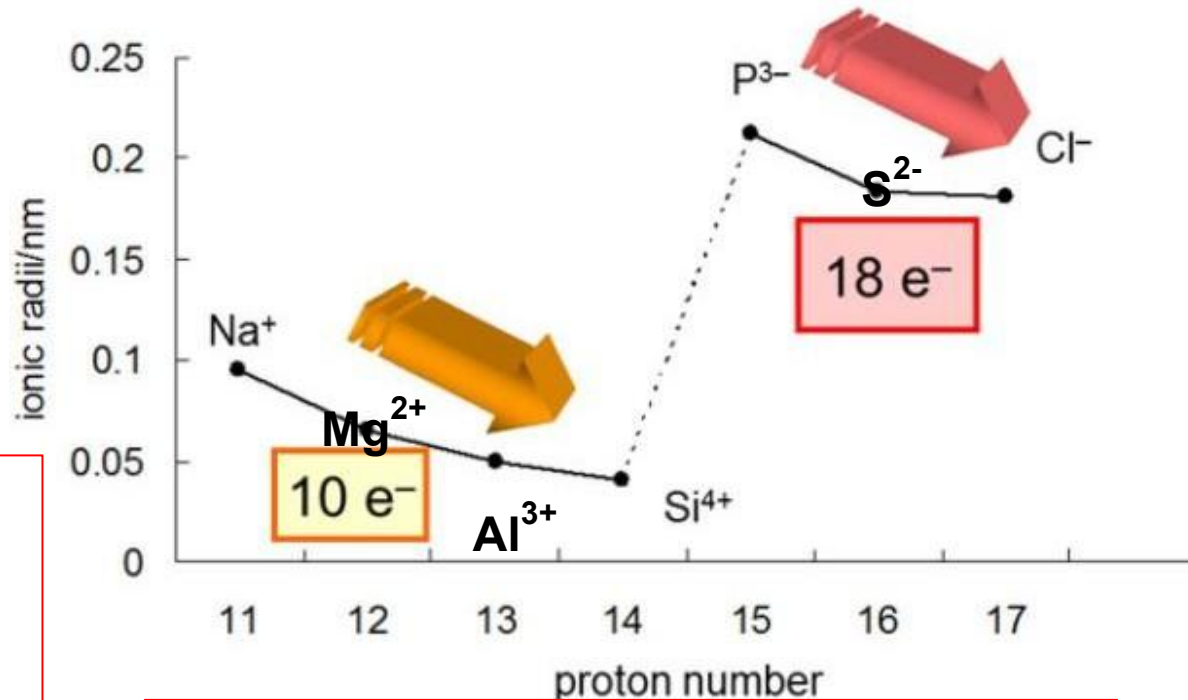
Cl⁻

1s² 2s² 2p⁶ 3s² 3p⁶

Principal
quantum
number, n = 2

Principal
quantum
number, n = 3

Graph of Ionic Radii vs Proton Number



- Na⁺, Mg²⁺, Al³⁺ and Si⁴⁺ ions are **isoelectronic cations**.
- As the proton number increases from Na⁺ to Si⁴⁺, the **effective nuclear charge increases**.
- The **attraction between the nucleus and valence electrons becomes stronger**.
- Thus, the ionic radius **decreases** from Na⁺ to Si⁴⁺.

- **There is a big jump** in radius from Si⁴⁺ ion to P³⁻ ion because of the **increase in the number of electron shells @ increase in the principal quantum number (from n=2 to n=3)**.
- As the number of shells filled with electrons increases, the **shielding effect increases**.
- **Weaker attraction** between the **nucleus** and the **valence electrons**.

- P³⁻, S²⁻ and Cl⁻ ions are **isoelectronic anions**.
- As the proton number increases from P³⁻ to Cl⁻, the **effective nuclear charge increases**.
- The **attraction between the nucleus and valence electrons increases**.
- Thus, the ionic radius of anions **decreases** from P³⁻ to Cl⁻.

All anions have larger radius than cations because **anions have one electron shell more than cations**, hence experience greater **shielding effect**.

Variation Of Ionic Radii Across Period 2

Proton Number	Ion	Electronic configuration	
3	Li ⁺	1s ²	Principal quantum number, n = 1
4	Be ²⁺	1s ²	
5	B ³⁺	1s ²	
6	C ⁴⁺	1s ²	
7	N ³⁻	1s ² 2s ² 2p ⁶	Principal quantum number, n = 2
8	O ²⁻	1s ² 2s ² 2p ⁶	
9	F ⁻	1s ² 2s ² 2p ⁶	

Example:

Explain the variation of ionic radii across period 2.

Ion	Li^+	Be^{2+}	B^{3+}	N^{3-}	O^{2-}	F^-
Radius (nm)	0.060	0.031	0.020	0.171	0.140	0.136

Guidelines:

To solve this question, you need to divide your answer into three parts.

- ✓ First : Compare ionic radius among isoelectronic cations, from Li^+ to B^{3+}
- ✓ Second: Big jump in ionic radius between cation B^{3+} and anion N^{3-}
- ✓ Third : Compare ionic radius for all isoelectronic anions from N^{3-} to F^-
- ✓ Fourth : Compare ionic radius between cations and anions.

ANSWER

Comparison of Cations

- Cations Li^+ , Be^{2+} and B^{3+} are **isoelectronic species** with **electronic configuration of $1s^2$** .
- From Li^+ , Be^{2+} to B^{3+} , as proton number increases, **effective nuclear charge also increases**, so the **attraction of nucleus towards the valence electrons becomes stronger**.
- Therefore, the **ionic radius decreases from Li^+ , Be^{2+} to B^{3+}** .

Big jump in ionic radius between cation B^{3+} and anion N^{3-}

- There is a **big jump in ionic radius from B^{3+} to N^{3-}** because of the increase in the number of electron shells (from $n = 1$ to $n = 2$).
- As the number of shells filled with electrons increases, the **shielding effect increases**.
- **Weaker attraction** between the **nucleus and the valence electrons**.

Comparison of Anions

- **Anions N^{3-} , O^{2-} and F^{-} are isoelectronic species with electronic configuration of $1s^2 2s^2 2p^6$.**
- **From N^{3-} , O^{2-} to F^{-} , as proton number increases, effective nuclear charge increases, so the attraction of nucleus towards the valence electrons becomes stronger.**
- **Therefore, the ionic radius decreases from N^{3-} , O^{2-} to F^{-} .**

Comparison of Cations and Anions

All anions (N^{3-} , O^{2-} and F^{-}) have larger radius than cations (Li^{+} , Be^{2+} and B^{3+}) because anions have one electron shell more than cations, hence experience greater shielding effect and weaker attraction between the nucleus and the valence electrons.

Review Questions

1. Determine the **period, group and block** for the following elements:
X: $1s^2 2s^2 2p^2$
Y: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
Z: $[\text{Ar}] 3d^5 4s^2$
2. Rank each set of ions in order of decreasing radii and explain.
 - (a) Ca^{2+} , Sr^{2+} , Mg^{2+}
 - (b) K^+ , S^{2-} , Cl^-

Question 1:

Answer

X: $1s^2 2s^2$

$2p^2$
Valence shell, $n = 2$

No. of valence electrons = 4

Subshell occupied by
highest energy electron = p

⇒ period 2

⇒ group 14

⇒ p block

Y: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

Valence shell, $n = 4$

No. of valence electron = 1

Subshell occupied by

highest energy electron = s ⇒ s-block

⇒ period 4

⇒ group 1

Question 1:



Z: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$

Valence shell, $n = 4 \Rightarrow$ period 4

Number of valence electrons = 7 $\Rightarrow$ group 7

Subshell occupied by
highest energy electron = d $\Rightarrow$ d-block

Question 2:



(a) Radii: $\text{Sr}^{2+} > \text{Ca}^{2+} > \text{Mg}^{2+}$

From Sr^{2+} , Ca^{2+} to Mg^{2+}

- Number of shells filled by electrons decreases.
- Shielding effect decreases.
- Strength of attraction between nucleus and valence electrons increases.
- Thus, ionic radius decreases.

(b) Radii: $\text{S}^{2-} > \text{Cl}^- > \text{K}^+$

S^{2-} , Cl^- and K^+ are isoelectronic species.

From S^{2-} , Cl^- to K^+

- Proton number increases
- Effective nuclear charge increases
- Strength of attraction between nucleus and valence electrons increases.
- Thus, ionic radius decreases.

FIRST AND SECOND IONISATION ENERGIES

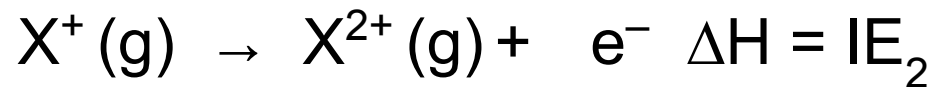
First Ionisation Energy, IE_1

The minimum energy required to **remove one mole of electrons** from **one mole of gaseous atoms to produce unipositive ions**.



Second Ionisation Energy, IE_2

The minimum energy required to **remove one mole of electrons** from **one mole of gaseous unipositive ions**.



- The magnitude of ionisation energy indicates how much energy is required to remove an electron.
- **The greater the ionisation energy, the more difficult to remove an electron.**
- In general, **smaller atoms have higher ionisation energy and vice-versa.**
- The same factors that influence atomic size also influence ionisation energy.

Example:

With reference to the table given below

Element	$_{11}\text{Na}$	$_{12}\text{Mg}$
Ionisation energy (kJ mol^{-1})	+496	+738

(a) Define **first ionisation energy**.

✓ The **minimum energy** required to **remove one mole of electrons** from **one mole of gaseous atoms to produce unipositive ions**.

(b) Write the **equation that represents the first ionisation energy of Na and Mg** respectively.



Example:

With reference to the table given below

Element	$_{11}\text{Na}$	$_{12}\text{Mg}$
Ionisation energy (kJ mol^{-1})	+496	+738

(c) Why is the **first ionisation energy of sodium lower than that of magnesium?** Na : $1s^2 2s^2 2p^6 3s^1$



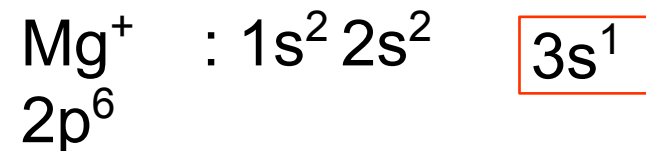
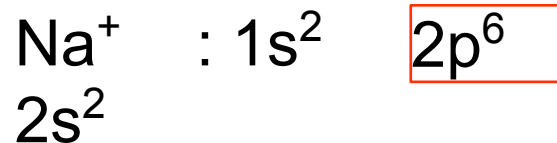
- ✓ Na has lower proton number than Mg, hence **lower effective nuclear charge** than Mg.
- ✓ The **attraction of nucleus towards the valence electrons** in Na is weaker as compared to Mg.
- ✓ Na has **larger atomic size** than Mg.
- ✓ Thus, **less energy** is required to remove the first electron in Na.
- ✓ As a result, Na has lower first ionisation energy.

Example:

With reference to the table given below

Element	$_{11}\text{Na}$	$_{12}\text{Mg}$
Ionisation energy (kJ mol ⁻¹)	+496	+738

(d) Which element has higher second ionisation energy?



- ✓ **Na has higher second ionisation energy than Mg.**
- ✓ **The second electron of Na is removed from the stable completely filled 2p orbital in Na⁺ which is closer to nucleus, hence more energy is required for the removal of electron.**
- ✓ **In Mg, the second electron is removed from the 3s orbital in Mg⁺ which is located further from the nucleus compared to Na⁺. The electron is held less tightly by the nucleus, hence easier to be removed.**

Variations In The First Ionisation Energy Across A Period And Down A Group

Group 1 Group 2

Increasing shielding effect

Increasing atomic radius

Decreasing IE_1

Li

152

Na

186

K

227

Rb

248

Cs

265

Be

112

Mg

160

Ca

197

Sr

215

Ba

222

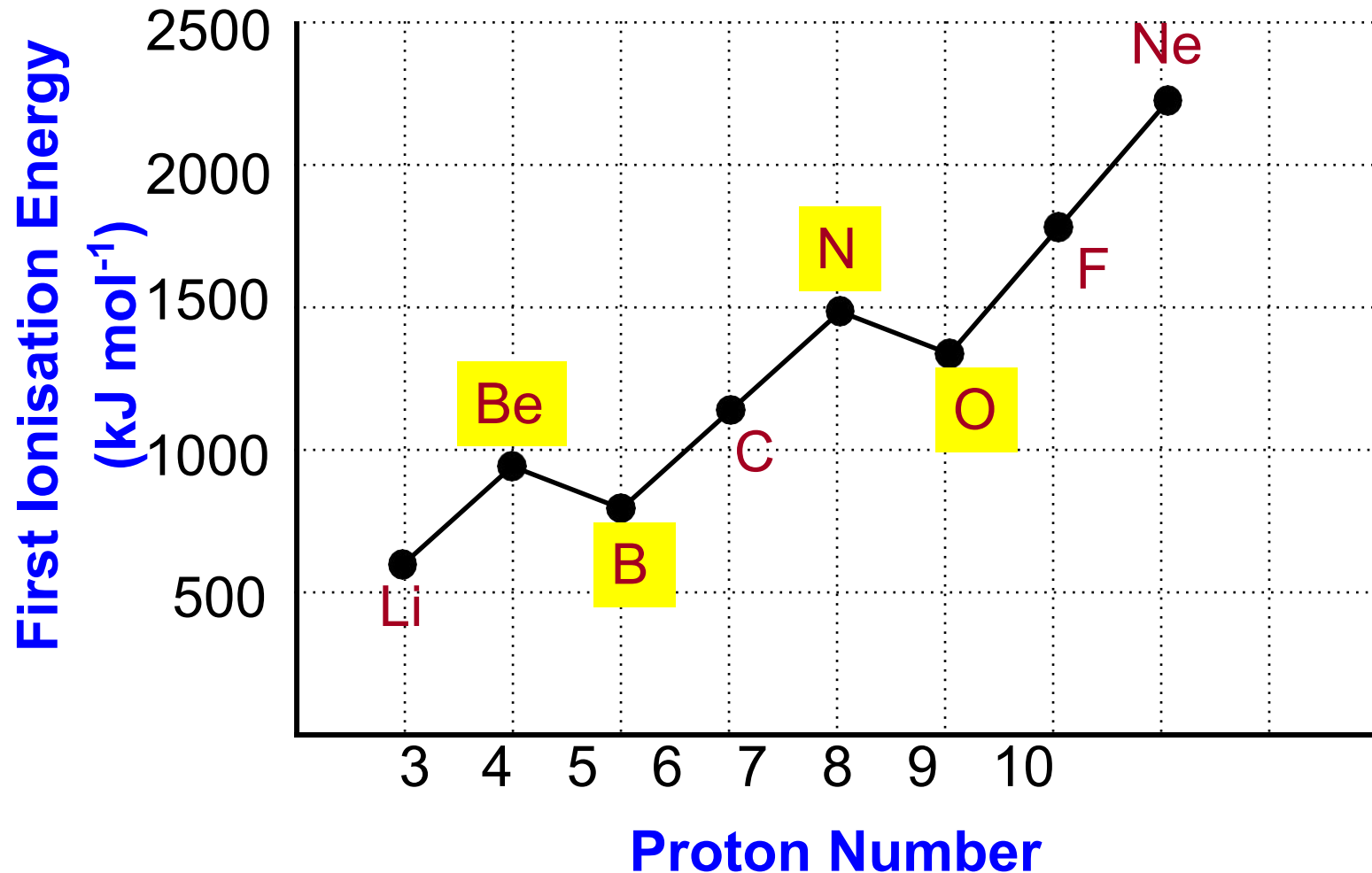
Going down a group:

- ✓ When number of electron shells or principal quantum number increases, shielding effect increases
- ✓ Attraction between nucleus and valence electrons becomes weaker.
- ✓ Atomic size increases.
- ✓ Less energy is required to remove the first valence electron
- ✓ First ionisation energy decreases.

Variation In First Ionisation Energy Across A Period

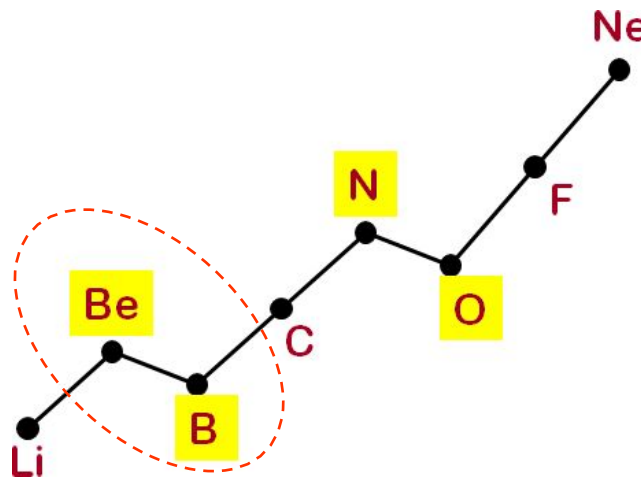
- ✓ When proton number increases, **effective nuclear charge** (Z_{eff}) increases
- ✓ The attraction between nucleus and valence electrons becomes stronger.
- ✓ Atomic size decreases.
- ✓ More energy is required to remove the first valence electron.
- ✓ First ionisation energy increases.
- ✓ However, there are slight irregularities in this trend.
- ✓ The irregularities occur between group 2 and group 13 and also between group 15 and group 16.

Variation In First Ionisation Energy Across Period 2



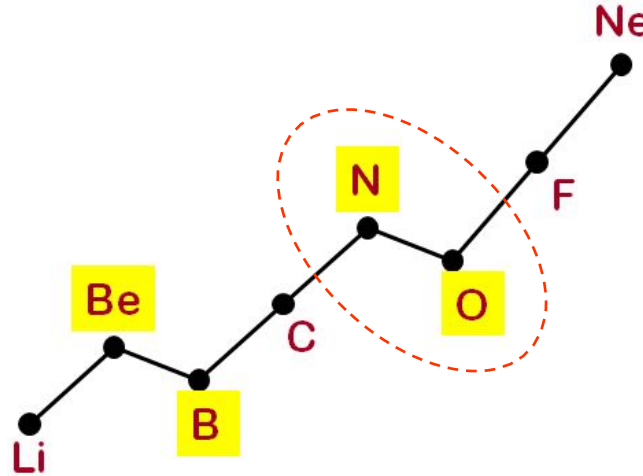
The anomalous behaviour can only be explained by inspecting the valence electronic configurations.

Irregularities Between Group 2 (Be) and Group 13 (B) In Period 2



- ✓ **Be** is found to have a **higher first ionisation energy** than **B**.
- ✓ This irregularity can be explained by the **stability of the completely filled orbital**.
- ${}_4\text{Be}: 1s^2 2s^2$ → First valence electrons of Be is removed from the **completely filled 2s orbital**.
- ${}_5\text{B}: 1s^2 2s^2 2p^1$ → First valence electrons of B is removed from the **partially filled 2p orbital**.
- ✓ **Completely filled 2s orbital is more stable than partially filled 2p orbital.**
Therefore, **more energy is required to remove the first valence electrons in Be atom than B atom**

Irregularities Between Group 15 (N) and Group 16 (O) In Period 2

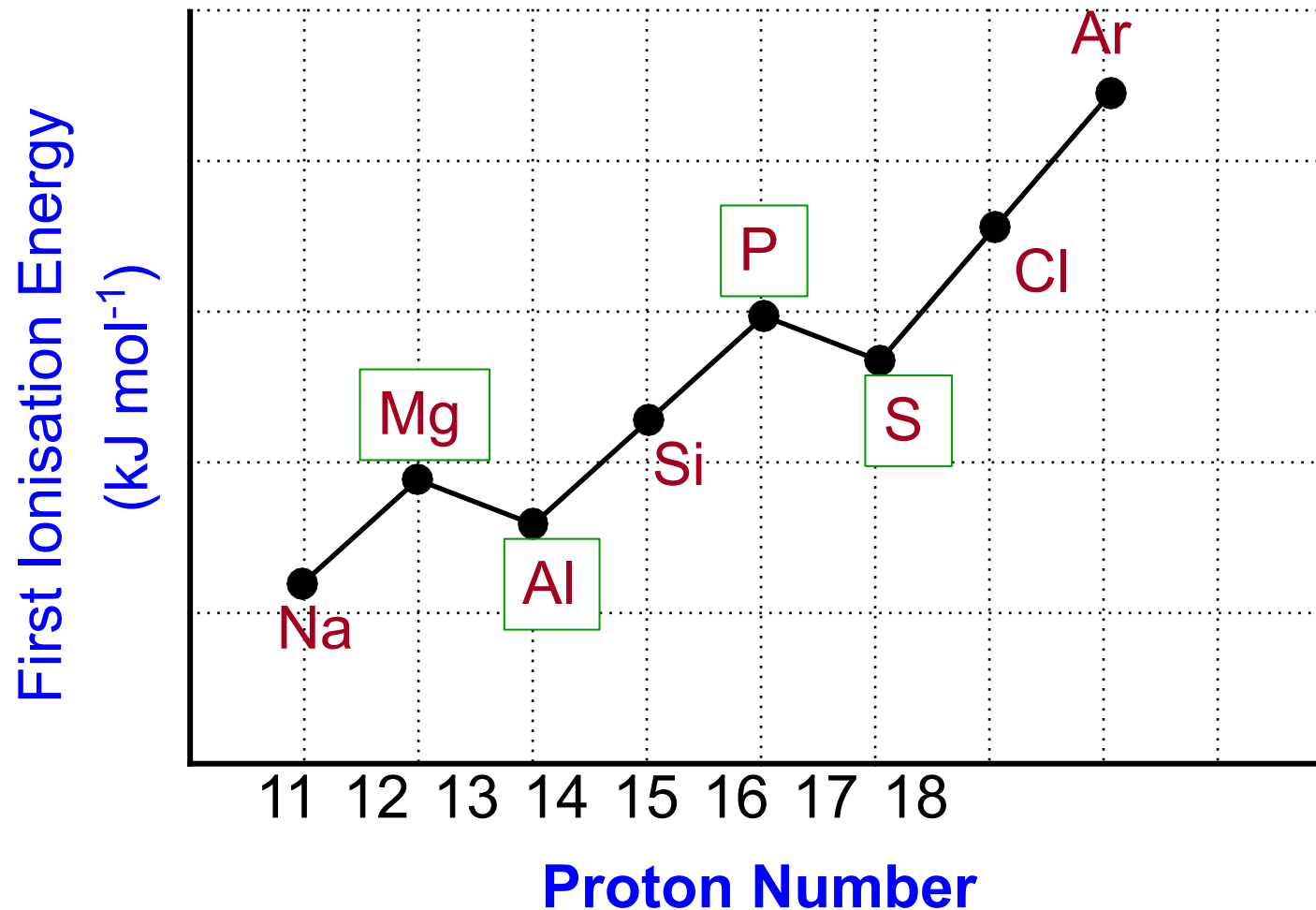


- ✓ N is found to have a **higher first ionisation energy** than O.
- ✓ This irregularity can be explained by **the stability of the half filled orbital**.
- ${}^7\text{N}: 1s^2 2s^2 2p^3$ → The first valence electrons of N is removed from the **half filled 2p orbital**.
- ${}^8\text{O}: 1s^2 2s^2 2p^4$ → The first valence electrons of O is removed from the **partially filled 2p orbital**.
- ✓ **Half-filled 2p orbital is more stable than partially filled 2p orbital**, therefore more energy is required to remove the **first valence electrons** in N atom than O atom.

Variation In First Ionisation Energy Across Period 3

Example:

Explain the variation of first ionisation energy across period 3





Answer

Across Period 3 from Na to Ar,

- ✓ When proton number increases, **effective nuclear charge** (Z_{eff}) increases
- ✓ Attraction between nucleus and valence electrons become stronger.
- ✓ Atomic size @ radius decreases.
- ✓ More energy required to remove the first valence electron.
- ✓ Therefore, generally the first ionisation energy increases.
- ✓ However, for **atoms whose valence electrons are in half-filled orbitals (P) or completely filled orbital (Mg)**, the ionisation energy is **higher than the expected value** due to the **extra stability of the orbitals**.

Irregularities In Period 3 Elements Between Group 2 & 13 and Group 15 & 16

$_{12}\text{Mg}: 1s^2 2s^2 2p^6 3s^2$ ✓ Mg has **completely filled 3s orbital**.

$_{13}\text{Al}: 1s^2 2s^2 2p^6 3s^2 3p^1$ ✓ Al has **partially filled 3p orbital**.

✓ More energy is required to **remove the first valence electrons in more stable completely filled 3s orbital** in **Mg** atom than **partially filled 3p orbital** in **Al** atom.

$_{15}\text{P}: 1s^2 2s^2 2p^6 3s^2 3p^3$ ✓ P has **half filled 3p orbital**.

$_{16}\text{S}: 1s^2 2s^2 2p^6 3s^2 3p^4$ ✓ S has **partially filled 3p orbital**.

✓ More energy is required to **remove the first valence electrons in more stable half filled 3p orbital** in **P** atom than **partially filled 3p orbital** in **S** atom.

Variation in Successive Ionisation Energies

✓ If an atom has n electrons, its successive ionisation energies show increasing trend:

$$IE_1 < IE_2 < IE_3 \dots < IE_n$$

Inner electrons

Group		first	second	third	fourth	fifth	sixth	seventh	
1	Li	520	7297	11810	-	-	-	-	Ionisation energy (kJ mol ⁻¹)
2	Be	900	1757	14840	21000	-	-	-	
13	B	800	2430	3659	25020	32810	-	-	
14	C	1086	2352	4619	6221	37800	47300	-	
15	N	1402	2857	4577	7473	9443	53250	64340	
16	O	1314	3391	5301	7468	10980	13320	71300	

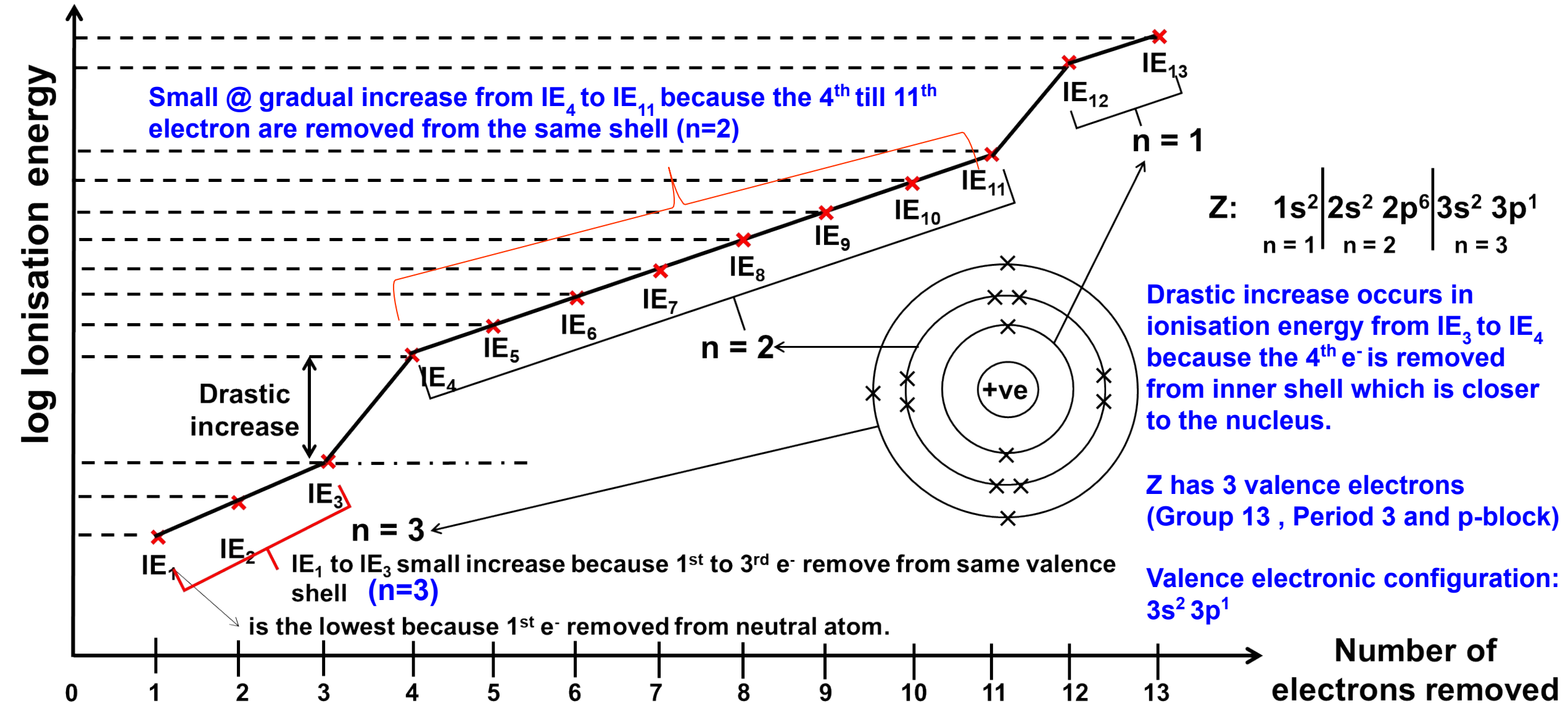
← Valence electrons →

✓ Successive IE values can be used to deduce the position of elements in the Periodic Table.

Variation in Successive Ionisation Energies

- ✓ **Successive ionisation energy will increase for each of the electron being removed.** This is because the **attraction of nucleus toward the remaining electrons becomes stronger when an electron is removed.** Thus, more energy is required to remove the next electron.
- ✓ **The successive ionisation energies increase gradually when the electrons are removed from the same shell @ valence shell, but increase drastically when the electrons are removed from an inner shell which is closer to nucleus.**
- ✓ **Inner electrons are much closer to the nucleus as compared to valence electrons, hence are held more tightly by nucleus.** Therefore, a drastic increase in the ionisation energy occurs when successive electrons are removed from different shells.
- ✓ **Analysing the data of the successive ionisation energy, IE will enable us to figure out the valence electronic configuration and the position of element in the Periodic table.**

Example: The following graph shows **all the successive energies** of an element Z.



Example:

An element X has the following successive ionisation energies, IE :

	IE ₁	IE ₂	IE ₃	IE ₄	IE ₅	IE ₆	IE ₇	IE ₈
Ionisation energy (kJ mol ⁻¹)	786	1580	3230	4360	16000	20000	23600	29100

i. Determine the group of X is located in the periodic table. Explain

$\frac{IE_2}{IE_1}$	$\frac{IE_3}{IE_2}$	$\frac{IE_4}{IE_3}$	$\frac{IE_5}{IE_4}$	$\frac{IE_6}{IE_5}$	$\frac{IE_7}{IE_6}$	$\frac{IE_8}{IE_7}$
$\frac{1580}{786}$ = 2.01	$\frac{3230}{1580}$ = 2.04	$\frac{4360}{3230}$ = 1.35	$\frac{16000}{4360}$ = 3.67	$\frac{20000}{16000}$ = 1.25	$\frac{23600}{20000}$ = 1.18	$\frac{29100}{23600}$ = 1.23

✓ **Group 14**

✓ **Drastic increase in ionisation energy from IE₄ to IE₅ @ $\frac{IE_5}{IE_4}$ highest ratio, more energy is required to remove the 5th electron.**

✓ **5th electron is removed from inner shell which is closer to nucleus.**

✓ **X has 4 valence electrons.**

ii. Write the valence electronic configuration of an atom of X.

✓ **Valence electronic configuration of X: ns² np²**

Example:

The first five ionisation energy of two elements are given below:

Determine the position of each element in the periodic table. Explain

Element	Ionisation Energy (kJ mol ⁻¹)				
	first	second	third	fourth	fifth
A	786	1540	3230	4360	16000
B	418	3070	4600	5860	7990

Element	IE2 IE1	IE3 IE2	IE4 IE3	IE5 IE4
A	1.96	2.10	1.35	3.67
B	7.34	1.50	1.27	1.36

Element A:

⇒ **Drastic increase** in ionisation energy from **IE₄ to IE₅** @ $\frac{IE_5}{IE_4}$ highest ratio indicates that **more energy is required** to remove the **5th electron**.

⇒ The **electron is removed** from **inner^{5th} shell** which is **close to nucleus**.

⇒ A has **4 valence electrons** (ns² np²)

⇒ **Group 14, p-block**.

Element B:

⇒ **Drastic increase** in ionisation energy from **IE₁ to IE₂** @ $\frac{IE_2}{IE_1}$ highest ratio indicates that **more energy is required** to remove the **2nd electron**.

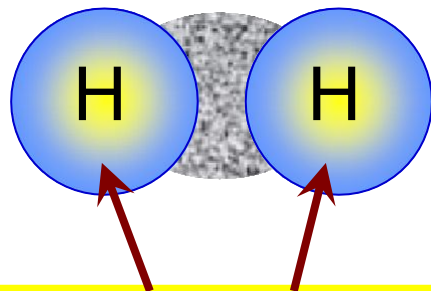
⇒ The **electron is removed** from **inner^{2nd} shell** which is **close to nucleus**.

⇒ A has **1 valence electrons** (ns¹)

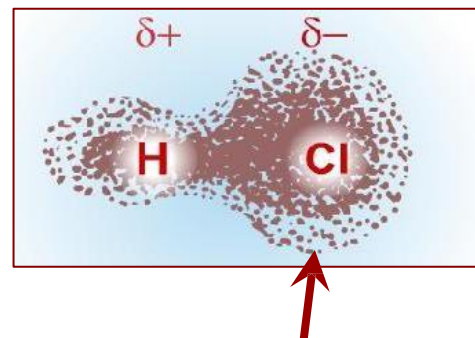
⇒ **Group 1, s-block**.

Electronegativity

- ✓ Electronegativity is the **ability or relative tendency of an atom to attract bonding electrons to itself** when chemically combined with another atom.
- ✓ The **greater the electronegativity** of an atom in a molecule, the **more strongly** it attracts the bonding electrons.



Same electronegativity



Cl is more electronegative

Variation In Electronegativity Of Elements

Across Period $\Rightarrow$ Electronegativity Increases

Down a Group $\Rightarrow$ Electronegativity Decreases

	1																	18
1	H 2.1	2											13	14	15	16	17	He
2	Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne
3	Na 0.9	Mg 1.2	3	4	5	6	7	8	9	10	11	12	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	Ar
4	K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr
5	Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 1.9	I 2.5	Xe
6	Cs 0.7	Ba 0.9	La 1.1	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2	Rn
7	Fr 0.7	Ra 0.9	Ac 1.1															

The most electronegative atom

The most electronegative atom

Across a Period

- ✓ Proton number increases.
- ✓ Effective nuclear charge, Z_{eff} increases.
- ✓ Atomic size @ atomic radius decreases.
- ✓ Ability @ relative tendency of atom to attract bonding electrons increases.
- ✓ Electronegativity increases.

Down a Group

- ✓ Number of electron shells increases
- ✓ Shielding effect increases.
- ✓ Atomic size @ atomic radius increases.
- ✓ Ability @ relative tendency of atom to attract bonding electrons decreases.
- ✓ Electronegativity decreases.

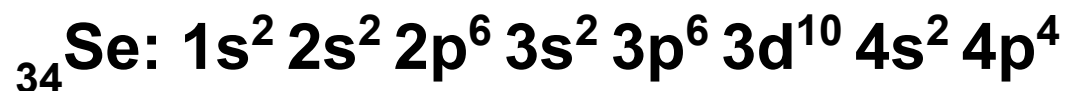
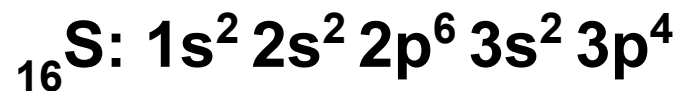
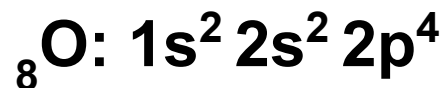
Down a Group $\Rightarrow$ Electronegativity Decreases

Across Period $\Rightarrow$ Electronegativity Increases

	1																	18
1	H 2.1	2											13	14	15	16	17	He
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3	Na 0.9	Mg 1.2	3	4	5	6	7	8	9	10	11	12	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	Ar
4	K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr
5	Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 1.9	I 2.5	Xe
6	Cs 0.7	Ba 0.9	La 1.1	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2	Rn
7	Fr 0.7	Ra 0.9	Ac 1.1															

Example:

Arrange oxygen, sulphur and selenium in the order of increasing electronegativity. Explain your answer.



Electronegativity: Se < S < O

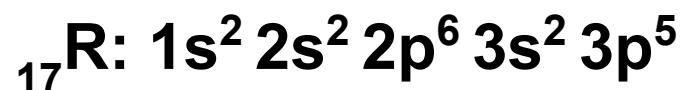
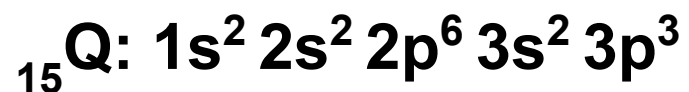
✓ Se, S and O are in the **same group @ Group 16** elements.

From Se, S to O,

- ✓ Number of electron shells or principal quantum number decreases
- ✓ Shielding effect decreases.
- ✓ Atomic size decreases.
- ✓ Ability @ relative tendency of atom to attract bonding electrons increases.
- ✓ Thus, electronegativity increases.

Example:

Given the proton number of elements **P**, **Q** and **R** is 12, 15 and 17 respectively. Arrange the elements **P**, **Q** and **R** in the order of increasing electronegativity. Give explanation for your arrangement.



Electronegativity: $P < Q < R$

- ✓ **P, Q and R are elements of the same period @ period 3.**
- ✓ **From P, Q to R.**
- ✓ **Proton number increases.**
- ✓ **Effective nuclear charge increases.**
- ✓ **Atomic size decreases.**
- ✓ **Ability @ relative tendency of atom to attract bonding electrons increases,**
- ✓ **Thus, electronegativity increases.**

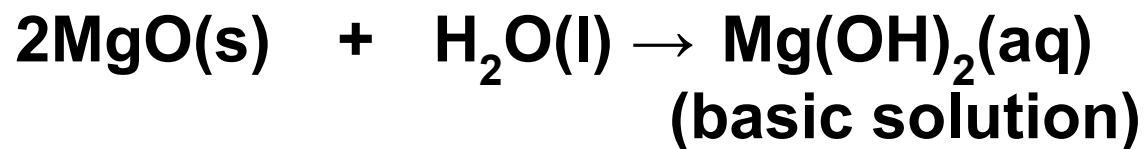
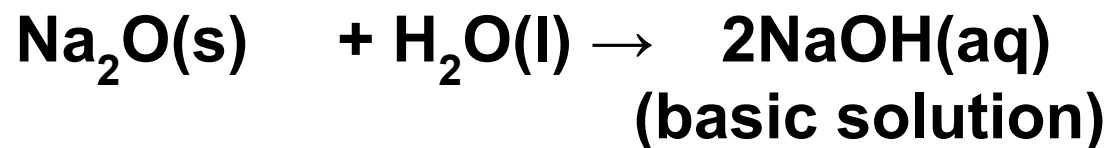
Acid-Base Character Of Oxides & Types Of Bonding Of Period 3 Elements

Element	Na	Mg	Al	Si	P	S	Cl
Oxide Formulae	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_6 @ P_4O_{10}	SO_2 @ SO_3	Cl_2O_7
Acid-base Properties	Basic	Basic	Amphoteric	Acidic	Acidic	Acidic	Acidic
Types of bonding	Ionic	Ionic	Ionic	Covalent	Covalent	Covalent	Covalent

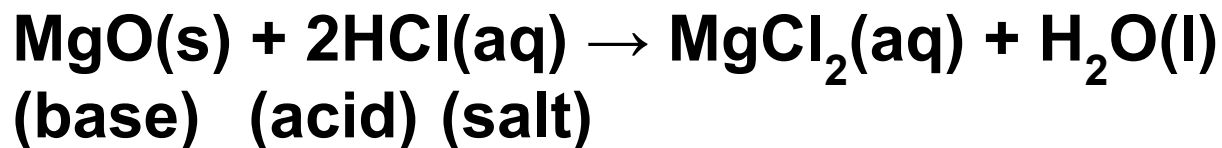
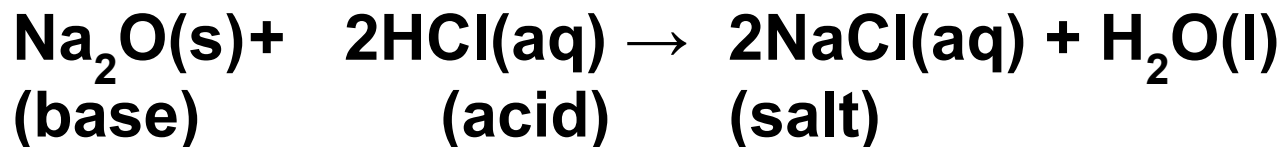
1) Basic oxide

Types of bonding : Ionic bond

- ✓ Sodium oxide and magnesium oxide are insoluble in water but react with water to form basic solutions.



- ✓ Sodium oxide and magnesium oxide are basic. They react with acid to form salt and water.

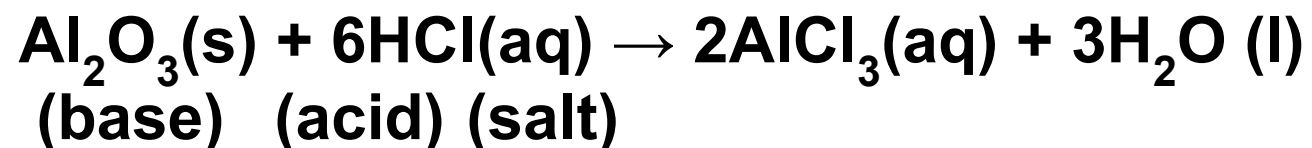


2. Amphoteric oxide

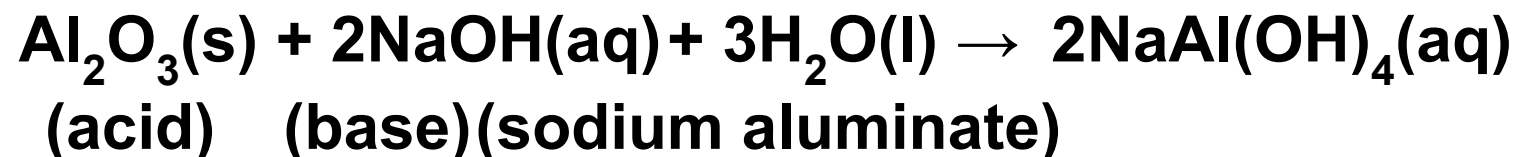
Types of bonding : Ionic bond

- ✓ **Aluminium oxide is insoluble in water but it is amphoteric oxide because it has properties of both acid and base.**
- ✓ **It can act as either an acid or a base.**

i. **When Al_2O_3 acts as base $\Rightarrow$ it reacts with acid**



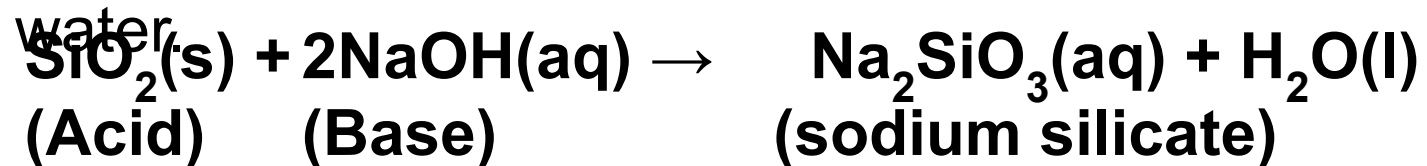
ii. **When Al_2O_3 acts as acid $\Rightarrow$ it reacts with base**



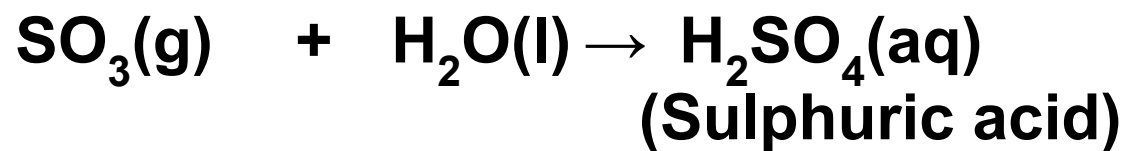
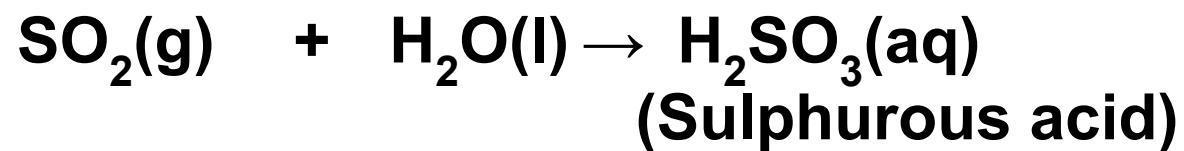
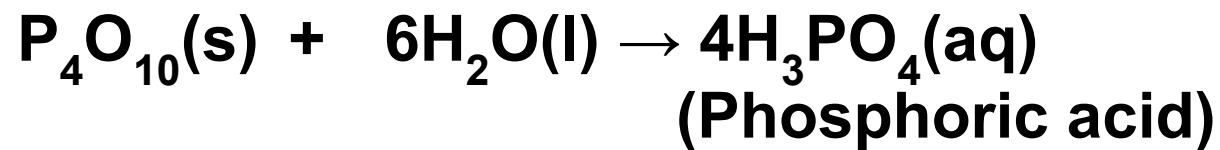
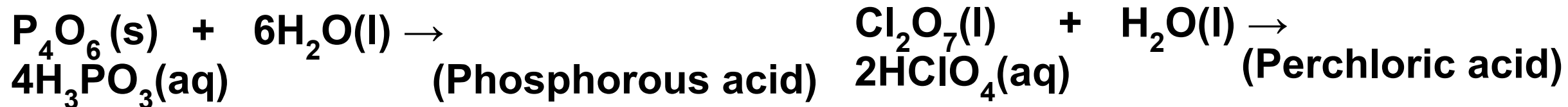
3. Acidic oxide

Types of bonding : Covalent bond

✓ **Silicon oxide is insoluble** in **water**, but **reacts** with **base** to form salt and water.



✓ The **oxides of nonmetals are acidic** because they react with water to **produce acidic solutions**:



An abstract graphic on the left side of the image, consisting of overlapping, wavy, translucent shapes in shades of green, yellow, orange, red, and purple, resembling a stylized flower or a series of waves.

THE END